

By Jeremiah Simon

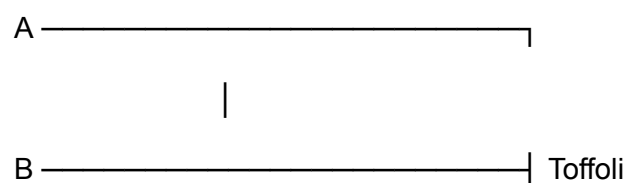
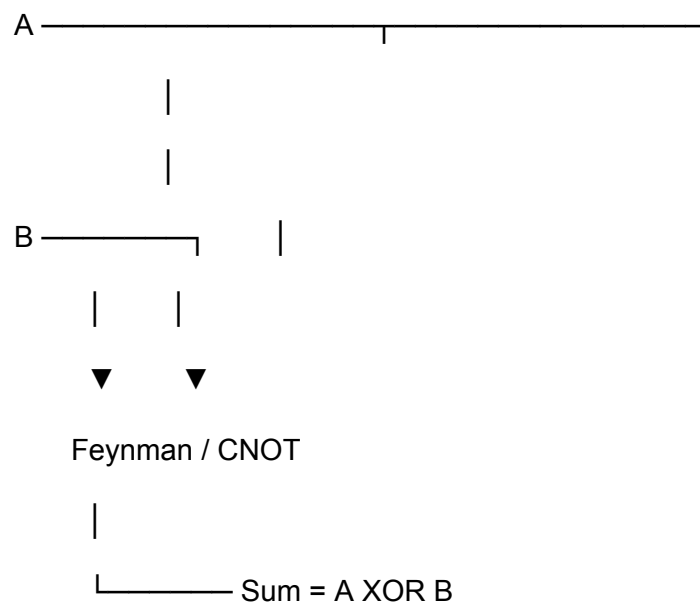
A conventional half adder produces:

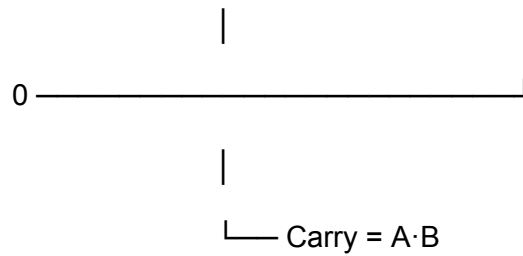
$$Sum = A \oplus B$$

$$Carry = A \cdot B$$

A reversible implementation can use a **Feynman gate** followed by a **Toffoli gate**.

Gate-Level Concept :





A common reversible realization is:

Feynman Gate

Use:

$$(A, B) \rightarrow (A, A \text{ XOR } B)$$

so the second output becomes the **Sum**.

Toffoli Gate

Use an ancilla input initialized to 0:

$$(A, B, 0) \rightarrow (A, B, A \cdot B)$$

so the third output becomes the **Carry**.

Thus:

$$\text{Sum} = A \text{ XOR } B$$

$$\text{Carry} = A \text{ AND } B$$

Half-Adder Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The 0 used by the Toffoli is an **ancilla/constant input**.